AP PRECALCULUS

Unit 1 • Lesson 1.8 • Guided notes

# Topic 1.8 • Rational Functions and Zeros

Guided notes • Fill in the blanks and work the examples in the boxes

**Learning goals**

- Find the real zeros of a rational function, checking the domain.
- Solve rational inequalities with a sign chart.

**Vocabulary.** zero, domain, excluded value, sign chart, test value, interval notation.

## 1. Zeros

**Key idea**

The real zeros of $r = \dfrac{p}{q}$ are the real zeros of ____ that are _____ (not zeros of $q$). Always _____ first.

**Example 1**

Find the zeros of $r(x) = \dfrac{x^2 - 3x}{x^2 - 9}$ and of $s(x) = \dfrac{x^2 - 4}{x + 1}$.

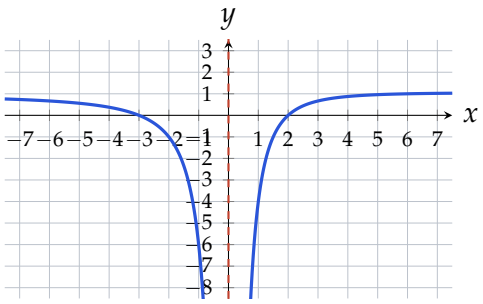
## 2. Sign charts

The sign of a rational function can change only at its _____ and at _____. Mark these on a number line, test _____, and read off the sign.

**Example 2 • Solve $r(x) \le 0$**

Solve $\dfrac{(x+1)(x-4)}{x-2} \le 0$.

### Example 3 • From a graph




The graph of $r(x) = \frac{(x+3)(x-2)}{x^2}$ is shown. Solve $r(x) > 0$.

**Common mistakes**

- Including excluded values in the solution of $\leq$ or $\geq$.
- Multiplying both sides of an inequality by an expression whose sign is unknown.

**Check your understanding**

1. Zeros of $\frac{x^2-1}{x+1}$: _____.  2. $\frac{x}{x-3} > 0$ on _____.